Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/20/2010
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNX09AO53A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor is roughly broken up with PBS and DLU performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work and computations at University of Hawaii, and RLA performing normal gravity experiments and computations at Washington University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	• The Requirements Definition Review (RDR) for this project was held in Cleveland during May 10 – 11, 2010. This review was successful and helped to advance synergies among the four ACME experiments. • The Science Requirements Document (SRD) for this project was extensively revised in preparation for the RDR. • The operational requirements, test matrix, and success criteria for this project were extensively revised in preparation for the RDR. These appear in the ACME science requirements document. • The presence of appreciable molecular oxygen at the location of maximum temperature has been observed and modeled in non-premixed oxygen-enhanced combustion processes. This work was published in one journal paper. This is significant for understanding soot formation in flames. • A paper considering kinetic and radiative extinction of spherical diffusion flames was prepared and submitted to a journal. This paper is important because an improved understanding of extinction can lead to improved combustor stability and efficiency. • Numerical simulations of sooting limits of spherical diffusion flames have been performed. This work has been published in one conference paper. • Two students completed their Ph.D. studies on this research.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: II. Flame structure effects on soot precursor kinetics resulting in soot-free flames." Combustion and Flame, in press, July 2010. Expected publication September 2010. , Jul-2010
Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: I. The presence of appreciable oxygen at the location of maximum temperature." Combustion and Flame 2009 Nov;156(11):2145–52. http://dx.doi.org/10.1016/j.combustflame.2009.07.009 , Nov-2009
Articles in Peer-reviewed Journals	Wang Q, Chao BH. "Kinetic and Radiative Extinctions of Spherical Diffusion Flames." Combustion Theory and Modeling, submitted, as of July 2010. , Jul-2010
Dissertations and Theses	Lecoustre VR. "Numerical Investigations of Gaseous Spherical Diffusion Flames." Ph.D. Dissertation, University of Maryland, Department of Mechanical Engineering, December, 2009. , Dec-2009
Dissertations and Theses	Skeen SA. "Oxygen-Enhanced Combustion: Theory and Applications." Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, August, 2009. , Aug-2009
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Axelbaum RL. "Numerical Simulations of Soot Kinetics in Spherical Diffusion Flames." Eastern States Meeting of the Combustion Institute, College Park, Maryland, May 2009. Eastern States Meeting of the Combustion Institute, College Park, Maryland, May 2009, 8 pp. , May-2009
Papers from Meeting Proceedings	Lecoustre VR, Moran CW, Sunderland PB, Chao BH, Axelbaum RL. "Experimental and Numerical Investigation of Extremely Weak Hydrogen Diffusion Flames." 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009, 12 pp. , May-2009
Papers from Meeting Proceedings	Lecoustre VR, Sunderland PB, Chao BH, Urban DL, Stocker DP, Axelbaum RL. "Numerical Simulations of Spherical Diffusion Flames at their Sooting Limits." 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 6th U.S. National Combustion Meeting, Ann Arbor, MI, May 2009. 9 pp. , May-2009